

# Experimental Design Concept Summary

## Key Concepts from Lectures 0-6 (CRD, RCBD, Factorial)

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# 1 Foundational Principles of Experimental Design

## 1.1 Fisher's Three Pillars

- **Replication:** Multiple experimental units per treatment to:
  - Estimate variability within treatments
  - Improve precision of treatment effect estimates
  - Enable valid statistical inference
  - **Biological replication** (different plants/animals/plots) drives inference
  - **Technical replication** (repeated measurements on same unit) improves precision but doesn't replace biological reps
- **Randomization:** Random assignment of treatments to experimental units to:
  - Eliminate systematic bias
  - Ensure independence of errors
  - Validate statistical tests (F-test, t-test)
  - Balance unknown confounding factors
- **Blocking:** Grouping similar experimental units to:
  - Control known sources of variation (spatial gradients, temporal effects)
  - Reduce experimental error (MS\_Error)
  - Increase power to detect treatment effects
  - Ensure fair comparison of treatments

## 1.2 Experiments vs. Observational Studies

- **Experimental study:** Researcher assigns treatments to units; can infer causation
- **Observational study:** Researcher observes without assigning treatments; limited causal inference
- **Risk:** Treating observational patterns as causal leads to poor recommendations

## 1.3 Experimental Unit vs. Observational Unit

- **Experimental unit:** Smallest unit to which a treatment is independently applied
- **Observational unit:** Unit on which measurements are taken
- **True replication:** Multiple experimental units per treatment
- **Pseudo-replication:** Multiple observations on the same experimental unit (technical replication)
- **Example:** Plants in same pot = pseudo-replication; plants in different pots = true replication

## 2 Completely Randomized Design (CRD)

### 2.1 When to Use CRD

- **Ideal for:** Controlled environments (growth chambers, labs, homogeneous fields)
- **Assumption:** All experimental units are homogeneous (no known gradients)
- **Not appropriate for:** Field trials with soil gradients, greenhouse bench effects, or any known heterogeneity

### 2.2 CRD Statistical Model

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$$

Where:

- $y_{ij}$  = observation  $j$  in treatment  $i$
- $\mu$  = overall mean
- $\tau_i$  = effect of treatment  $i$  (fixed effect)
- $\varepsilon_{ij} \sim N(0, \sigma^2)$  = random error

**Alternative parameterization:**

$$y_{ij} = \mu_i + \varepsilon_{ij}$$

where  $\mu_i$  = mean of treatment  $i$

### 2.3 CRD ANOVA Table

| Source    | df         | Sum of Squares | Mean Square | F-ratio                 |
|-----------|------------|----------------|-------------|-------------------------|
| Treatment | $t - 1$    | $SS_{Trt}$     | $MS_{Trt}$  | $\frac{MS_{Trt}}{MS_E}$ |
| Error     | $t(r - 1)$ | $SS_E$         | $MS_E$      |                         |
| Total     | $tr - 1$   | $SS_{Total}$   |             |                         |

Where:

- $t$  = number of treatments
- $r$  = number of replicates per treatment
- Total observations =  $t \times r$

## 2.4 Degrees of Freedom Formulas (CRD)

- **df\_Treatment** =  $t - 1$  (number of treatments minus 1)
- **df\_Error** =  $N - t = t(r - 1)$  (total observations minus number of treatments)
- **df\_Total** =  $N - 1 = tr - 1$  (total observations minus 1)

**Example:** 4 treatments  $\times$  6 reps = 24 total observations

- **df\_Treatment** =  $4 - 1 = 3$
- **df\_Error** =  $24 - 4 = 20$
- **df\_Total** =  $24 - 1 = 23$

## 3 ANOVA Assumptions and Diagnostics

### 3.1 Three Critical Assumptions

#### 3.1.1 1. Normality of Residuals

- **What:** Errors ( $\varepsilon_{ij}$ ) follow a normal distribution
- **Not** the raw data, but the **residuals** from the model
- **Check with:** Q-Q plot, Shapiro-Wilk test
- **Robustness:** ANOVA is robust to mild violations with balanced designs and  $n > 30$  per group

#### 3.1.2 2. Homogeneity of Variance (Homoscedasticity)

- **What:** All treatment groups have equal error variance ( $\sigma^2$ )
- **Check with:** Residuals vs. Fitted plot, Levene's test, Bartlett's test
- **Violation:** Heteroscedasticity biases F-test, makes comparisons unfair

#### 3.1.3 3. Independence of Errors

- **What:** No correlation between observations
- **Ensured by:** Proper randomization and experimental design
- **Common violations:**
  - Spatial correlation (adjacent plots)
  - Temporal correlation (repeated measures)
  - Cluster effects (animals in same pen)

### 3.2 Diagnostic Plots

#### 3.2.1 Q-Q Plot (Quantile-Quantile)

- **Purpose:** Check normality of residuals
- **Good:** Points fall on diagonal line
- **Bad:** Systematic deviation from line
  - S-curve = skewness
  - Fatter/thinner ends = heavy/light tails

### 3.2.2 Residuals vs. Fitted Values

- **Purpose:** Check homoscedasticity and detect patterns
- **Good:** Random scatter around zero, constant spread
- **Bad:**
  - Funnel shape = heteroscedasticity
  - Curved pattern = non-linearity
  - Outliers visible

### 3.2.3 Scale-Location Plot

- **Purpose:** Check homogeneity of variance
- **Good:** Horizontal line with random scatter
- **Bad:** Increasing/decreasing trend

## 4 Randomized Complete Block Design (RCBD)

### 4.1 When to Use RCBD

- Use when experimental units are **heterogeneous** but variation can be grouped
- Block on **known nuisance factors** you're not testing (spatial gradients, time, litter, batch)
- Each block contains **one complete set** of all treatments
- Randomization occurs **within blocks**

### 4.2 Common Blocking Factors

- **Field trials:** Soil gradient, slope, drainage, fertility zones
- **Greenhouse:** Bench position, proximity to vents/doors
- **Orchards:** Tree age, row position, rootstock
- **Animal studies:** Litter, age, initial weight
- **Time effects:** Planting date, harvest day, processing batch

### 4.3 RCBD Statistical Model

$$y_{ij} = \mu + \beta_j + \tau_i + \varepsilon_{ij}$$

Where:

- $y_{ij}$  = response for treatment  $i$  in block  $j$
- $\mu$  = overall mean
- $\beta_j$  = effect of block  $j$  (removes heterogeneity)
- $\tau_i$  = effect of treatment  $i$  (what we're testing)
- $\varepsilon_{ij} \sim N(0, \sigma^2)$  = random error (within-block variability)

**Key assumption: Additivity** – block and treatment effects are additive (no block  $\times$  treatment interaction)

### 4.4 RCBD ANOVA Table

| Source    | df               | Sum of Squares | Mean Square  | F-ratio                   |
|-----------|------------------|----------------|--------------|---------------------------|
| Block     | $b - 1$          | $SS_{Block}$   | $MS_{Block}$ | $\frac{MS_{Block}}{MS_E}$ |
| Treatment | $t - 1$          | $SS_{Trt}$     | $MS_{Trt}$   | $\frac{MS_{Trt}}{MS_E}$   |
| Error     | $(b - 1)(t - 1)$ | $SS_E$         | $MS_E$       |                           |
| Total     | $bt - 1$         | $SS_{Total}$   |              |                           |

Where:

- $b$  = number of blocks
- $t$  = number of treatments
- Total observations =  $b \times t$



## 4.5 Degrees of Freedom Formulas (RCBD)

- **df\_Block** =  $b - 1$  (number of blocks minus 1)
- **df\_Treatment** =  $t - 1$  (number of treatments minus 1)
- **df\_Error** =  $(b - 1)(t - 1)$  (interaction df; used as error in RCBD)
- **df\_Total** =  $bt - 1$  (total observations minus 1)

**Example:** 5 blocks  $\times$  4 treatments = 20 total observations

- $\text{df\_Block} = 5 - 1 = 4$
- $\text{df\_Treatment} = 4 - 1 = 3$
- $\text{df\_Error} = (5-1) \times (4-1) = 12$
- $\text{df\_Total} = 20 - 1 = 19$

## 4.6 Additivity Assumption

- **Tukey's Test for Additivity:** Tests if block  $\times$  treatment interaction is present
- **If significant:** Additivity violated; RCBD model invalid
- **Fix:** Interpret the results by block; use a different design to account for interactions

## 4.7 CRD vs. RCBD Comparison

| Feature              | CRD                                                               | RCBD                                       |
|----------------------|-------------------------------------------------------------------|--------------------------------------------|
| <b>Units</b>         | Homogeneous                                                       | Heterogeneous (grouped into blocks)*       |
| <b>Randomization</b> | Completely random                                                 | Restricted (within blocks)                 |
| <b>Error</b>         | $MS_E = \frac{SS_E}{t(r-1)}$                                      | $MS_E = \frac{SS_E}{(b-1)(t-1)}$           |
| <b>Precision</b>     | If heterogeneous E.U.,<br>Lower (includes between-unit variation) | Higher (blocks remove variation)           |
| <b>Complexity</b>    | Simpler                                                           | More complex (requires blocking structure) |

\* E.U. withing blocks should be as homogeneous as possible; over-blocking can reduce power

## 5 Factorial Designs

### 5.1 Factorial Notation and Structure

- **2×2 factorial:** 2 factors, each with 2 levels → 4 treatment combinations
- **3×3 factorial:** 2 factors, each with 3 levels → 9 treatment combinations
- **2×4 factorial:** 2 factors (2 levels, 4 levels) → 8 treatment combinations
- **2×2×2 factorial:** 3 factors, each with 2 levels → 8 treatment combinations

### 5.2 Advantages of Factorial Designs

- Test **multiple factors simultaneously** in one experiment
- More **efficient/powerful** than one-factor-at-a-time (OFAT) experiments
- Detect **interactions** between factors (cannot be detected in OFAT!)
- Broader scope of inference (generalize across factor combinations)

### 5.3 Main Effects

- **Main effect of Factor A:** Average effect of Factor A **across all levels of Factor B**
- Compares marginal means (averaged over other factors)
- Answers: “On average, does this factor matter?”

**Example:** Main effect of Irrigation = (Mean yield under High Irrigation) - (Mean yield under Low Irrigation), averaged over all Nitrogen levels

### 5.4 Interactions

- **Interaction:** Effect of Factor A **depends on** the level of Factor B
- Main effects are **not additive**
- Simple effects (A at each level of B) differ

**Example:** Nitrogen increases yield by 200 kg/ha under low irrigation but 1000 kg/ha under high irrigation → **Irrigation × Nitrogen interaction**

### 5.5 Factorial Statistical Model (2-Factor CRD)

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$

Where:

- $y_{ijk}$  = response for Factor A level  $i$ , Factor B level  $j$ , replicate  $k$
- $\mu$  = overall mean
- $\alpha_i$  = main effect of Factor A at level  $i$
- $\beta_j$  = main effect of Factor B at level  $j$
- $(\alpha\beta)_{ij}$  = interaction effect between A and B
- $\varepsilon_{ijk} \sim N(0, \sigma^2)$  = random error

## 5.6 Factorial ANOVA Table (2-Factor CRD)

| Source       | df               | F-ratio                |
|--------------|------------------|------------------------|
| Factor A     | $a - 1$          | $\frac{MS_A}{MS_E}$    |
| Factor B     | $b - 1$          | $\frac{MS_B}{MS_E}$    |
| A $\times$ B | $(a - 1)(b - 1)$ | $\frac{MS_{AB}}{MS_E}$ |
| Error        | $ab(r - 1)$      |                        |
| Total        | $abr - 1$        |                        |

Where:

- $a$  = number of levels of Factor A
- $b$  = number of levels of Factor B
- $r$  = number of replicates per combination
- Total observations =  $a \times b \times r$

## 5.7 Degrees of Freedom Formulas (2-Factor Factorial in CRD)

- $df\_A = a - 1$  (levels of Factor A minus 1)
- $df\_B = b - 1$  (levels of Factor B minus 1)
- $df\_AB = (a - 1)(b - 1)$  (interaction df)
- $df\_Error = ab(r - 1)$  (total obs minus number of combinations)
- $df\_Total = abr - 1$  (total observations minus 1)

**Example:** 2 levels of Irrigation  $\times$  3 levels of Nitrogen  $\times$  4 reps = 24 total observations

- $df\_Irrigation = 2 - 1 = 1$
- $df\_Nitrogen = 3 - 1 = 2$
- $df\_Interaction = (2-1) \times (3-1) = 2$
- $df\_Error = 2 \times 3 \times (4-1) = 18$
- $df\_Total = 24 - 1 = 23$

## 5.8 Interpreting Interaction Plots

- **No interaction:** Lines are parallel
- **Interaction present:** Lines cross or diverge
- **When significant interaction exists:**
  - Main effects alone are **misleading**
  - Must examine **simple effects** (effect of A at each level of B)
  - Report and interpret interaction, not just main effects

## 5.9 Sum of Squares: Type I, II, and III

### 5.9.1 Why Different Types Matter

When calculating ANOVA, the **order of terms** and **handling of unbalanced data** affects how sum of squares (SS) are partitioned. Three types exist:

### 5.9.2 Type I SS (Sequential)

- **Calculation:** Each factor's SS is calculated **after** adjusting for factors already in the model
- **Order dependent:** SS for Factor B depends on whether Factor A is entered first
- **Default in:** Base R `anova()` function
- **Use when:**
  - Data is **balanced** (equal sample sizes)
  - You have a simple (**hierarchical** or **nested**) model structure
  - Order of factors has scientific meaning

**Example order:**

1.  $SS(A)$  — variation explained by A alone
2.  $SS(B|A)$  — additional variation explained by B after accounting for A
3.  $SS(AB|A,B)$  — additional variation explained by interaction after A and B

### 5.9.3 Type II SS (Partial)

- **Calculation:** Each main effect tested **after** all other main effects but **not** interactions
- **Order independent** for main effects
- **Use when:**
  - Data is **unbalanced**
  - **No significant interactions** are present
  - You want to test main effects independent of each other

**Example order:**

1.  $SS(A|B)$  — variation for A after adjusting for B (but not AB)
2.  $SS(B|A)$  — variation for B after adjusting for A (but not AB)
3.  $SS(AB|A,B)$  — interaction after both main effects

**Important:** Type II assumes **no interaction**; if interaction is significant, Type II for main effects is questionable.

### 5.9.4 Type III SS (Marginal)

- **Calculation:** Each factor tested **after** adjusting for **all other factors** including interactions
- **Order independent:** Results same regardless of model entry order
- **Default in:** SAS, SPSS, `car::Anova(..., type = "III")` in R
- **Use when:**
  - Data is **unbalanced**
  - Interactions may be **present**
  - You want **marginal** effects (averaged over other factors)
  - Publishing in journals that expect Type III (common in agriculture/biology)

**Example order:**

1.  $SS(A|B, AB)$  — variation for A after adjusting for B and AB interaction
2.  $SS(B|A, AB)$  — variation for B after adjusting for A and AB interaction
3.  $SS(AB|A, B)$  — interaction after both main effects

**Critical requirement:** Use `contrasts = c("contr.sum", "contr.sum")` in R for proper Type III SS calculation (sum-to-zero constraints).

### 5.9.5 Comparison Table

| Feature                         | Type I (Sequential)              | Type II (Partial)            | Type III (Marginal)                      |
|---------------------------------|----------------------------------|------------------------------|------------------------------------------|
| <b>Order dependent</b>          | Yes                              | No                           | No                                       |
| <b>Adjusts for interactions</b> | No (sequential)                  | No (main effects only)       | Yes                                      |
| <b>Best for balanced data</b>   | Yes (all types give same result) | Yes                          | Yes                                      |
| <b>Best for unbalanced data</b> | No                               | Only if no interaction       | <b>Yes</b>                               |
| <b>Default in R</b>             | <code>anova()</code>             | <code>car::Anova(...)</code> | <code>car::Anova(..., type="III")</code> |
| <b>Default in SAS/SPSS</b>      | No                               | No                           | <b>Yes</b>                               |
| <b>Requires sum contrasts</b>   | No                               | No                           | <b>Yes (in R)</b>                        |

### 5.9.6 When to Use Each Type

Use **Type I** when:

- Balanced design (all types equivalent)
- Hierarchical/nested model with meaningful order
- Simple one-way ANOVA (only one factor)

Use **Type II** when:

- Unbalanced design **and** no significant interaction
- You want to test main effects after controlling for each other
- More powerful than Type III when no interaction exists

Use **Type III** when (most common in agriculture/biology):

- Unbalanced design
- Factorial design with potential interactions
- Publishing in journals requiring Type III (common standard)
- Want results comparable to SAS/SPSS
- Safest choice when unsure about interactions

### 5.9.7 Practical Example

**Scenario:** 2×2 factorial (Irrigation × Nitrogen) with unbalanced data

- Low Irrigation: n = 10 (N0), n = 15 (N100)
- High Irrigation: n = 12 (N0), n = 8 (N100)

**Type I** (order matters):

```
anova(lm(yield ~ irrigation + nitrogen + irrigation:nitrogen))  
# SS changes if you swap order to nitrogen + irrigation
```

**Type II** (assumes no interaction):

```
car::Anova(lm(yield ~ irrigation + nitrogen + irrigation:nitrogen))  
# Main effects tested without interaction adjustment
```

**Type III** (marginal effects, safest):

```
model <- lm(yield ~ irrigation * nitrogen,  
           contrasts = list(irrigation = "contr.sum", nitrogen = "contr.sum"))  
car::Anova(model, type = "III")  
# Each effect tested after adjusting for all others
```

### 5.9.8 Key Takeaway

- **Balanced designs:** All three types give **identical results**
- **Unbalanced designs:** Type III is the **safest choice** and most widely accepted
- **Always specify** which type you used in publications
- In R, use `car::Anova(..., type = "III")` with sum contrasts for factorial designs with unbalanced data

## 6 Multiple Comparisons and Post-Hoc Tests

### 6.1 When to Use Post-Hoc Tests

- **After** rejecting the null hypothesis in ANOVA (significant F-test)
- ANOVA tells us “at least one mean differs” but not **which ones**
- Post-hoc tests perform pairwise comparisons while controlling Type I error

### 6.2 Common Multiple Comparison Procedures

#### 6.2.1 Tukey's Honest Significant Difference (HSD)

- **Use:** Compare **all pairwise** treatment means
- **Controls:** Familywise error rate (FWER) at  $\alpha$
- **Conservative:** Balances Type I and Type II error
- **Best for:** Equal sample sizes (balanced designs)

#### 6.2.2 Dunnett's Test

- **Use:** Compare **treatment means to a control** only
- **More powerful** than Tukey when only control comparisons matter
- **Example:** Compare new fertilizers to standard practice

### 6.3 Planned Contrasts vs. Post-Hoc

- **Planned contrasts:** Hypothesis specified **before** data collection; more powerful
- **Post-hoc:** Exploratory comparisons **after** seeing data; requires error rate adjustment

### 6.4 Contrasts

- **Contrast:** Linear combination of treatment means with coefficients summing to zero
- **Orthogonal contrasts:** Two contrasts are orthogonal if their coefficients satisfy specific conditions
- **Advantage:** Can test specific scientific hypotheses (e.g., control vs. treatment average)



## 7 Key Statistical Concepts

### 7.1 Hypothesis Testing in ANOVA

#### 7.1.1 CRD Hypotheses

- $H_0 : \mu_1 = \mu_2 = \dots = \mu_t$  (all treatment means are equal)
- $H_a : \text{At least one } \mu_i \text{ differs}$

#### 7.1.2 RCBD Hypotheses (for treatments)

- $H_0 : \tau_1 = \tau_2 = \dots = \tau_t = 0$  (no treatment effects)
- $H_a : \text{At least one } \tau_i \neq 0$

#### 7.1.3 Factorial Hypotheses

For main effect A:

- $H_0 : \alpha_1 = \alpha_2 = \dots = \alpha_a = 0$
- $H_a : \text{At least one } \alpha_i \neq 0$

For interaction A×B:

- $H_0 : (\alpha\beta)_{ij} = 0 \text{ for all } i, j$
- $H_a : \text{At least one } (\alpha\beta)_{ij} \neq 0$

### 7.2 Type I and Type II Errors

- **Type I error ( $\alpha$ ):** Reject true null hypothesis (false positive)
- **Type II error ( $\beta$ ):** Fail to reject false null hypothesis (false negative)
- **Power** =  $1 - \beta$  = Probability of detecting a true effect

### 7.3 P-values and Significance

- **P-value:** Probability of observing data as extreme as ours if  $H_0$  is true
- **Significance level:** Typically  $\alpha = 0.05$
- **Interpretation:**
  - $p < 0.05$ : Reject  $H_0$ ; evidence of treatment effect
  - $p \geq 0.05$ : Fail to reject  $H_0$ ; insufficient evidence

## 7.4 Effect Size vs. Statistical Significance

- **Statistical significance** (p-value) depends on sample size and variability
- **Effect size** measures practical importance (e.g., mean difference in yield)
- **Large sample** can make trivial effects “significant”
- **Always report** both p-value and effect size (mean differences, confidence intervals)

## 8 Practical Considerations

### 8.1 Sample Size and Power

- **Larger sample** → greater power to detect effects
- **More replicates** → smaller standard errors, narrower confidence intervals
- **Blocking** → removes variability, increases power without increasing sample size

### 8.2 Confounding

- **Confounding:** When treatment effect cannot be separated from another factor
- **Example:** Testing preservative only in morning shift (temperature confounded with treatment)
- **Prevention:** Randomization and proper experimental design

### 8.3 Randomization Methods

- Use software (R, FieldHub) for reproducible randomization
- **Set a seed** for reproducibility
- Document randomization procedure
- Avoid systematic patterns (alphabetical, spatial convenience)

### 8.4 When to Transform Data

- **Reason:** Stabilize variance, improve normality
- **Common transformations:**
  - Log: Right-skewed data, multiplicative effects
  - Square root: Count data (Poisson)
  - Arcsine square root: Proportions/percentages
- **Caution:** Interpret results on transformed scale or back-transform

## 9 Summary Tables

### 9.1 Design Selection Guide

| Condition                                        | Recommended Design                     |
|--------------------------------------------------|----------------------------------------|
| Homogeneous units, 1 factor                      | CRD                                    |
| Heterogeneous units, 1 factor,<br>known gradient | RCBD                                   |
| 2+ factors, test interactions                    | Factorial (CRD or RCBD structure)      |
| Factors hard to change                           | Split-plot (not covered yet)           |
| Limited resources, many factors                  | Fractional factorial (not covered yet) |

### 9.2 Degrees of Freedom Quick Reference

| Design                         | df_Treatment          | df_Block | df_Error         | df_Total  |
|--------------------------------|-----------------------|----------|------------------|-----------|
| <b>CRD</b>                     | $t - 1$               | -        | $t(r - 1)$       | $tr - 1$  |
| <b>RCBD</b>                    | $t - 1$               | $b - 1$  | $(b - 1)(t - 1)$ | $bt - 1$  |
| <b>2-Factor</b><br><b>CRD</b>  | See interaction table | -        | $ab(r - 1)$      | $abr - 1$ |
| <b>2-Factor</b><br><b>RCBD</b> | See interaction table | $b - 1$  | Complex          | $abt - 1$ |

### 9.3 Common Mistakes to Avoid

1. **Confusing experimental unit with observational unit** → pseudo-replication
2. **Ignoring blocking when heterogeneity exists** → inflated error, low power
3. **Using CRD when RCBD is needed** → missing opportunity to reduce error
4. **Over-blocking** (blocks too small) → lose df for error, reduce power
5. **Interpreting main effects when interaction is significant** → misleading conclusions
6. **Not checking assumptions** → invalid p-values and confidence intervals
7. **P-hacking**: Running multiple tests without correction
8. **Treating observational data as experimental** → invalid causal claims

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### End of Summary

*This document covers core concepts from Lectures 0-6 (Introduction, CRD, RCBD, Factorials). Use this as a study guide for understanding experimental design principles, ANOVA interpretation, and proper statistical analysis.*